

# PoliMemeDecode: Humor That Speaks Politics

CUET CSE FEST 2025 — DATATHON

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## 1 Team Information

Team Name: Team CopyPaste

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Table 1: Team members and corresponding information

## 2 Problem Statement

Social media has given rise to the evolution of memes, acting as a good source of conveying humor through messages that are often entertaining and in some cases by promoting satire. Memes have also managed to touch different fields, grabbing the attention of audiences holding varying tastes. Countries like Bangladesh where different political parties and figures exist, each vying for attention from the public have become the front face of meme content. For a human, coming to fathom the **context and meaning** of such memes becomes very feasible. Nevertheless, for a machine, **extracting the context and purpose** from a meme, which is often a very unorganized and messy collection of images and texts overlapping each other, fighting to highlight themselves can often turn out to be much more of a challenge.

The task in this Datathon is to **train and build a ML/DL model** that will allow us to classify memes into two categories which are as follows:

- Political: Refers to any meme whose context is correlated with any political body, conveying messages, commentary or satire.
- Non Political: Refers to any sort of meme that is not connected to any political reference, figure or party, rather is more relatable on a daily life basis or targeting some non political field.

## 3 Dataset Overview

The challenge uses a labeled image dataset curated for binary classification of political vs. non-political content. The dataset contains 2,860 annotated training images along with 330 unlabeled

test images reserved for evaluation. The images represent diverse meme and media content which makes the classification task realistic and relevant to applications such as automated content moderation, media analytics and political communication filtering. All the training and test samples are provided in a unified image directory with corresponding labels in CSV format supporting efficient data ingestion for machine learning workflows. Before developing the model, statistical analysis was performed to assess class imbalance, file quality and structural consistency of the dataset.

There is a pretty significant class imbalance in the training data where one class has more than twice the number of samples as the other. The ratio of imbalance of non political to political memes is 2.35 : 1. This imbalance had to be considered when training classification models to prevent bias. The distribution of class types are as follows: NonPolitical has 2007 images and Political has 853 images.

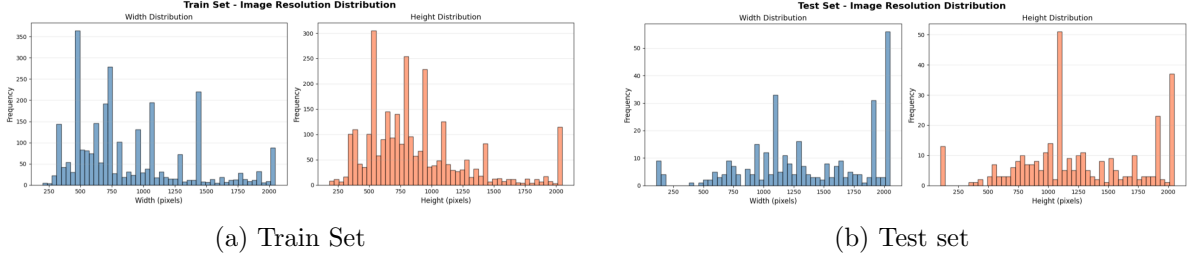


Figure 1: Resolution Distribution

The training set images range from  $200 \times 180$  to  $2048 \times 2048$  while the test set spans a slightly wider lower bound of  $110 \times 103$  up to the same maximum of  $2048 \times 2048$ . This shows that although both sets share a high-resolution upper limit, the test set contains more low-resolution images. Analyzing these resolution extremes is important because image size directly influences the model’s ability to capture visual features consistently. Understanding this variability helps ensure proper preprocessing, resizing strategies and robustness of the model across diverse image qualities.

## 4 Methodology

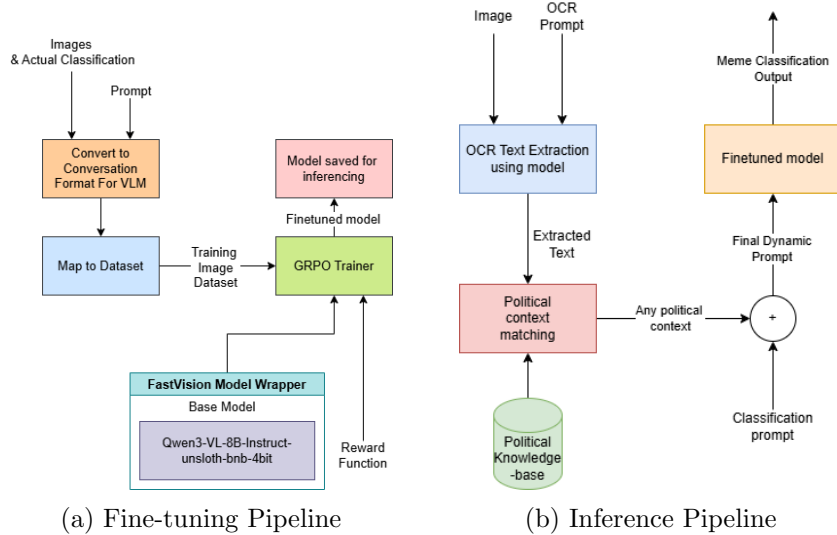


Figure 2: Overview of Model Training and Inference Pipelines

### 4.1 Model Selection

Meme classification into political or non-political classes requires a model to understand not only the images shown but understand the context behind it and relate it with the text in the meme.

Examining the dataset revealed that it is multi-lingual (Bengali and English) in nature and consists of images showing political and non-political figures. There were many instances where a meme may have a political figure but the accompanying text makes the meme have a different meaning, making it non-political. As a result, this classification task requires a model that can understand the context, image and text and properly link them to understand the main meaning behind the meme.

Understanding this challenge, we decided to work with Vision Language Models due to their multimodal nature and ability to understand images and texts at the same time. Keeping the GPU limitation in mind, we used 4 bit quantized VLM models offered by Unsloth. To choose the best model we experimented with the following models:

- Qwen3 VL-8B
- Qwen2.5 VL-7B
- Gemma-3-4B
- Qwen3 VL-32B (did not fit into free tier GPU, so was discarded)

Each model was used for inference using the same prompt and outputs were submitted for evaluation. Out of all base models, Qwen3 VL-8B-Instruct (4 bit quantized) gave the best results on the public leaderboard. Since Qwen models are better at multilingual tasks, we chose Qwen3 VL-8B-Instruct as our base model for finetuning.

## 4.2 Data Preprocessing

The data preprocessing pipeline involved several systematic steps to prepare the meme dataset for fine-tuning. First, 2,800 samples were loaded from a CSV file containing image filenames and binary labels (Political/NonPolitical) using pandas. Each meme image was then loaded as a PIL image object, resized to a standardized 512×512 pixel resolution (required for Qwen3), and converted to RGB format to ensure consistency across the dataset. The data was structured in a conversation format, where each sample contained a role-based prompt with an image placeholder and the classification instruction text, along with the corresponding ground-truth label. Finally, the tokenizer’s chat template was applied to format the conversational structure properly, generating prompts that included the required vision-token sequence:

```
<|vision_start|><|image_pad|><|vision_end|>
```

along with the instruction text asking the model to classify the meme as either *political* or *non-political* without providing reasoning.

## 4.3 Model Architecture and Configuration

The model was configured with LoRA (Low-Rank Adaptation) with rank 16 and alpha 16, training only 0.50% of total parameters (43.6M out of 8.8B). Crucially, vision layers were frozen (fine-tune\_vision.layers=False) while language layers, attention modules, and MLP modules remained trainable. We experimented with making the vision layer trainable as well but this resulted in the model performing slightly worse, indicating that the model was experiencing catastrophic forgetting. Thus it was decided to train with the vision layers freed, which resulted in better performance.

## 4.4 GRPO Trainer Implementation

We employed Qwen’s GRPO (Group Relative Policy Optimization) trainer, a reinforcement learning approach with the following configuration:

- Learning rate: 5e-6 with cosine scheduling
- 8-bit AdamW optimizer for memory efficiency
- Batch size of 1 with 4 gradient accumulation steps
- DR-GRPO loss type with sequence-level importance sampling

## 4.5 Domain-Aware Reward Function

A custom balanced reward function was implemented for class imbalance using the DISCO approach. Political memes (minority class) received weight 1.47 with  $\pm 2.94$  rewards/penalties, while non-political memes (majority class) received weight 0.88 with  $\pm 1.76$  rewards/penalties. This logarithmic scaling encouraged the model to learn the underrepresented political class more effectively.

The model was trained on all images for 0.5 epochs (700 steps) and then saved to a personal hugging face account for inference later on.

## 4.6 Inference

The fine-tuned model’s predictions were examined and it was observed that it was still struggling to understand several Bengali words especially when in different variations or in banglish. Moreover, since the model was unaware of recently emerging political figures, memes mentioning their name and conveying a political tone were being inferred as being non-political. To tackle this pitfall without introducing new images into the dataset, we implemented a RAG agent with a political knowledgebase. Via this approach, we can make the model aware of up to date developments without requiring further training.

The inference pipeline works in two passes. First, the vision model works like an OCR system and extracts all visible text from the meme. Any Bengali text is translated into English. This extracted text becomes the query. The political knowledge base contains information about Bangladeshi political figures, parties, events, and their aliases. This solves a major issue in political meme classification: the difficulty of recognizing many types of Bangladeshi political references (recent and old), especially when memes include multilingual text, spelling variations, transliterations, or slang. All entries are pre-encoded using the "sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2" sentence transformer, an open source (under the Apache License 2.0) multilingual model that produces 384-dimensional embeddings and works well for cross-lingual semantic similarity. The query text is encoded into the same vector space and cosine similarity is computed between the query and all knowledge base embeddings. Higher scores mean stronger semantic similarity. The system retrieves the top 10 entries with similarity above 0.3. These entries, along with their scores and descriptions, are formatted into a context string that is integrated with the entire prompt which tells the vision model which political entities or concepts are likely present. So for each meme the model is able to classify them with more up to date context.

This RAG design has several advantages. It handles multilingual and fuzzy matching, so variations of names like "Sheikh Hasina", or "shaikh hasina" still point to the same entity. It is efficient because the knowledge base embeddings are computed once and similarity checks are cheap vector operations. It is also interpretable since similarity scores show what was matched and how confidently. The process separates entity detection from final classification, making both parts more reliable. The threshold and top-k filtering help reduce noise and keep only the most relevant political context, which improves precision and recall.

## 5 Results and Analysis

Our inference resulted in all 330 samples processed successfully with no failures, indicating a stable and reliable pipeline. The final classification distribution is nearly even across the two categories, with **166 samples labeled as NonPolitical** and **164 samples labeled as Political**. This corresponds to proportions of **50.30% NonPolitical** and **49.70% Political** classification. We ran inference on the same code multiple times and got exactly the same result, ensuring that our predictions are consistent and reproducible. Our rag based inference with cosine similarity on the finetuned model achieved a Macro F1 score of **0.88509** on 30% of the test set in the public leader board and a Macro F1 score of **0.85274** on the remaining 70% of the test set in the private leader board. Compared to our initial base model (Public Macro F1 : 0.85895 and Private Macro F1: 0.78098) this was an increase of **3.04%** in Public Macro F1 and an increase of **9.19%** in Private Macro F1 score.